

1 The Zika virus is a pathogen that can infect human cells.

(a) Fig. 1.1 is a drawing of the structure of a Zika virus.

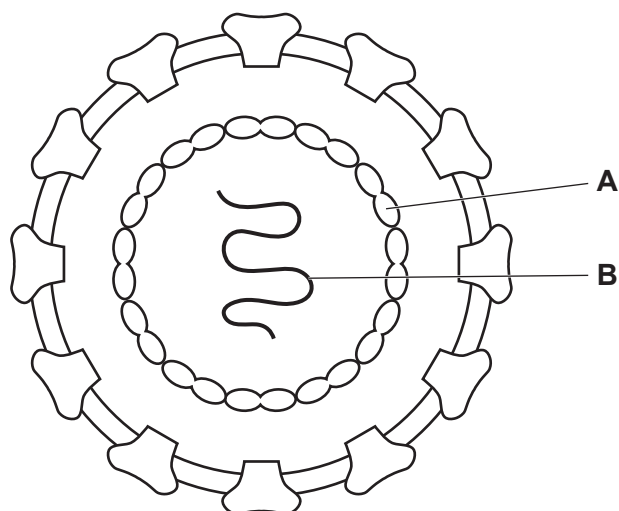


Fig. 1.1

(i) In Fig. 1.1, structure **A** is made of protein.

Name structure **A**.

..... [1]

(ii) In Fig. 1.1, structure **B** is a single-stranded molecule.

Suggest the name of structure **B**.

..... [1]

- (b) Fig. 1.2 is a transmission electron micrograph of human kidney cells infected with Zika viruses.

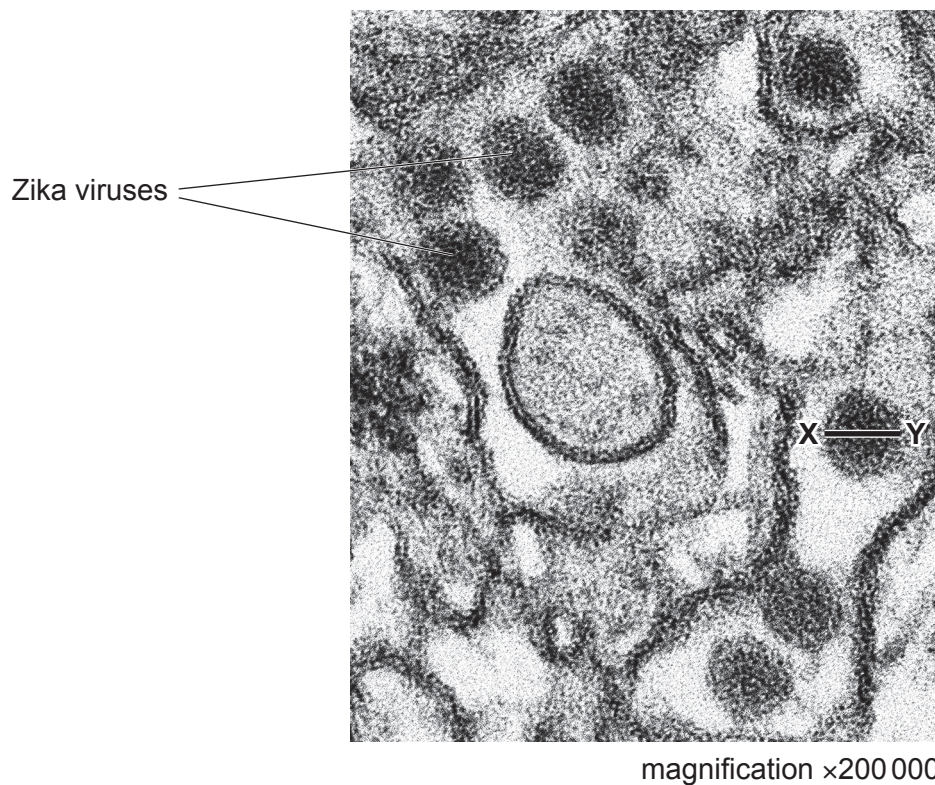


Fig. 1.2

- (i) Calculate the actual diameter of a Zika virus using the line **X–Y** in Fig. 1.2.

Show your working.

Give your answer in nanometres (nm).

diameter = nm [2]

- (ii) A magnification of $\times 200\,000$ cannot be achieved by a light microscope. The resolution of a transmission electron microscope is also higher than a light microscope.

Describe what is meant by the resolution of a microscope.

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 [1]

- (c) The vector for Zika virus is the mosquito *Aedes aegypti*. The mosquito feeds on the blood of an infected person and transmits the virus to another person when it feeds again.

Describe the similarities and differences between the transmission of Zika virus disease and the transmission of malaria.

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..... [3]

- (d) Zika virus vaccines have been developed by scientists.

One of the vaccines contains small proteins from the Zika virus.

- (i) Explain how giving this vaccine to a person can lead to the development of long-term immunity against Zika virus disease.

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- (ii) Explain how a vaccination programme may limit the spread of Zika virus disease through a population.

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[Total: 14]